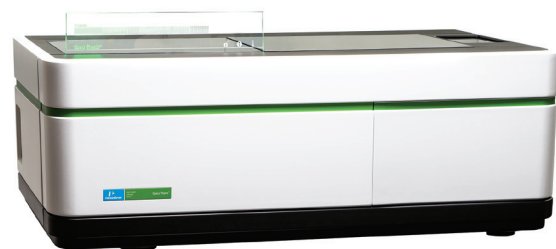


Cellular Imaging and Analysis

Opera Phenix™ High Content Screening System



Hardware

Detection

- Up to four parallel confocal detection channels
- Innovative, proprietary Synchrony™ Optics to minimize crosstalk during simultaneous acquisition
- True point confocal, microlens enhanced wide view Nipkow spinning disk
- 16 bit sCMOS 4.6 Megapixel (2160x2160), 6.5 μm pixel size
- Synchronized multi-color excitation and detection
- Transmitted light capabilities (brightfield, digital phase contrast)
- High speed, high resolution linear drive scanning stage, 50 nm resolution, 1 μm repeatability
- z-stage resolution 50 nm
- Measurement time: 5 min/384-well plate for a 4 color experiment and one field, depending on instrument configuration and assay

Excitation

- Configurations for either four or five lasers (see table)
- LED light source for transmission mode

Color	Wavelength [nm]	Standard Power [mW]	High Power [mW]
Violet/UV	405	50	200
	Laser 1: 375 Laser 2: 425	Laser 1: 16 Laser 2: 100	Laser 1: 50 Laser 2: 100
Blue	488	60	150
Green/ Yellow	561	50	100
Red	640	40	140

Objective lenses

- Six position objective turret for automated change
- Proprietary, fully automated high NA water immersion objectives
- Default objective: 20x air long WD, NA 0.4
- 20x water immersion lens, NA 1.0
- 40x water immersion lens, NA 1.1
- 63x water immersion lens, NA 1.15
- 1.25x air long WD, NA 0.03
- 5x air long WD, NA 0.16
- 10x air long WD, NA 0.3
- 40x air long WD, NA 0.6

Environmental control unit

- Temperature control: 37 °- 42 °C +/- 1 °C
- CO₂ control: 1-10 % +/- 0.5%

Physical dimensions

- Width: 134 cm / 52.8 in
- Depth: 65 cm / 25.6 in
- Height: 45 cm / 17.7 in

Robotic automation

- Cell::explorer™ robotic automation platform for fully automated cellular and high content screening
- Integrates with plate::handler™ robotic workstation and plate::works™ scheduling software
- Compatible with third party automation systems
- Add stackers, incubators and liquid handling

Computer

- Image analysis workstation: Intel® 2x Hexa Core™ (server processor), ~ 32 GB RAM
- Hard drive: ~ 2.2TB, RAID10 - additional hard drives available
- Operating system: Microsoft® Windows® 7, 64 bit
- Network interface: Gigabit Ethernet

Computer specifications subject to change without notice

Harmony® High Content Imaging and Analysis Software

Image Acquisition

- Computer controlled laser and excitation power
- Laser-based, high speed autofocus in every field
- Four-color parallel, sequential or mixed exposure image acquisition
- Digital phase contrast imaging
- Acquire and analyze simultaneously, and visualize data during acquisition
- Variable plate formats following SBS standard (6, 24, 48, 96, 384, 1536-well), user-defined formats and slides (in slide holder)
- Compatible with a wide range of microtiter and nanotiter plates from various manufacturers
- Full choice of number and locations of fields, focus height, volume imaging, plate format and type
- Multi-field imaging acquisition enabling full coverage of each well
- Z-sectioning for identifying the z-plane of interest for screening in various planes
- Individual plane or Maximum Projection Analysis
- Automated flat field correction
- Capabilities for time course experiments

Image analysis

- Ready-made turnkey applications
- Fully automated cell segmentation with automatic or manual parameter tuning
- Easy and intuitive creation and modification of analysis sequences
- PhenoLOGIC™ machine learning plug-in available, enabling users to teach the software to create tailored algorithms
- Texture analysis capabilities

Data Management

- Scalable database
- Search via sortable tree
- Stores all metadata automatically (assay layout, instrument setting, analysis results)
- Stores additional experiment information (user-defined keywords, annotations)
- Allows for image storage on a remote server
- Export results automatically or in batches into Columbus™ Image Data Storage and Analysis System so that you can access, re-analyze, store, and share your image data
- Export results to TIBCO Spotfire® for interactive data aggregation and visualization
- Drag and drop images into Volocity® 3D Image Analysis Software for 3D visualization and analysis.
- Export images and results to third party solutions
- Accessible from office installations of the Harmony software for data inspection, analysis and result export

Recommended Configurations

	Basic	Standard	Live Cell FRET	Screening
Camera (confocal) Lasers	1	2	4	4
375/425 nm	-	-	●	-
405 nm	●	●	-	●
488 nm	●	●	●	●
561 nm	●	●	●	●
640 nm	●	●	●	●

System Options				
Environmental Control	○	○	●	○
Water Immersion	○	●	●	●
Transmitted Light	●	●	●	●
Robotics	○	○	○	○

Imaging Modes				
Minimal Crosstalk Imaging	●	●	●	●
Ratiometric FRET	●	●	●	●
3D Imaging	●	●	●	●
Multi-color Fluorescence Imaging	●	●	●	●
Multi-color Simultaneous Confocal Imaging	-	●	●	●
Brightfield and Digital Phase Contrast	●	●	●	●
Fluorescence Widefield Imaging	●	●	●	●

Key: ● Included ○ Option - Unavailable

Note: Additional configurations available - please ask our specialist team for details.

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